Product, Quotient, and Trigonometric Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A differentiable function $f(t)$ is represented by the graph below. What is the total change in $f(t)$ from $t = 2$ to $t = 5$?

The total change in a function $f(t)$ from $t = a$ to $t = b$ is given by the definite integral $\int_a^b f'(t)\,dt$. Alternatively, if the graph represents the function $f(t)$ itself, the total change is simply $f(b) - f(a)$. From the graph, we can observe that $f(2) = 1$ and $f(5) = 2$. Therefore, the total change is $f(5) - f(2) = 2 - 1 = 1$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the function $f(x) = 3x^2 - 6x + 5$. What is the instantaneous rate of change of $f(x)$ at $x = 2$?

The instantaneous rate of change of a function at a point is given by its derivative evaluated at that point. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(3x^2 - 6x + 5) = 6x - 6$. Next, evaluate $f'(x)$ at $x = 2$: $f'(2) = 6(2) - 6 = 12 - 6 = 6$. Therefore, the instantaneous rate of change of $f(x)$ at $x = 2$ is 6.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A conical tank has height 15 ft and radius 7 ft at the top (radius:height ratio constant as it drains). Water drains at 6 ft^3/min. Find $\frac{dh}{dt}$ (in ft/min) when the water depth is 13 ft.

Since $r = \frac{7}{15}h$, $V = \frac{1}{3}\pi\left(\frac{7}{15}\right)^2 h^3$. Differentiating and solving for $\frac{dh}{dt}$ at $h = 13$ gives $-\frac{1350}{8281}/\pi$ ft/min.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Product rule for $\frac{d}{dx}[f(x)g(x)]$:.

Final Answer: $f'g + fg'$. Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, Product rule: $f'g + fg'$ Therefore, the verified correct option is $f'g + fg'$.

5 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{4x+2}$.

Final Answer: $\frac{1}{2(2x+1)^2}$. Quotient rule: $f'(x) = \frac{1\cdot(4x+2) - x\cdot 4}{(4x+2)^2} = \frac{1}{2(2x+1)^2}$.

6 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use the product rule to find $f'(x)$ if $f(x) = x^4(3x - 4)$.

Final Answer: $x^3(15x - 16)$. Product rule: $f'(x) = 4x^3(3x - 4) + x^4(3) = x^3(15x - 16)$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumA 13-foot ladder leans against a vertical wall. The base of the ladder slides away from the wall at a constant rate of 2 ft/s. How fast is the top of the ladder sliding down the wall when the base is 5 feet from the wall?

Let x be the distance from the base to the wall and y be the height on the wall. By the Pythagorean theorem: $x^2 + y^2 = 13^2 = 169$ At the instant $x = 5$: $y = \sqrt{169 - 25} = \sqrt{144} = 12$ ft. Differentiating implicitly with respect to time t: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow x\frac{dx}{dt} + y\frac{dy}{dt} = 0$ Substitute $x = 5$, $y = 12$, and $\frac{dx}{dt} = 2$ ft/s.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · AdvancedLet R be the region bounded by $y = \sqrt{x}$, $y = 2$, and the y-axis. Set up, but do not evaluate, an integral for the volume generated when R is revolved about the vertical line $x = -1$.

1. The region has boundaries $x = y^2$ (right boundary), $x = 0$ (left boundary / y-axis), and y runs from $y = 0$ to $y = 2$. 2. The axis of rotation is the vertical line $x = -1$. Slicing perpendicular to the axis of rotation requires integration with respect to y (horizontal washers). 3. Determine outer and inner radii:Outer radius (distance from $x = -1$ to furthest curve $x = y^2$): $R(y) = y^2 - (-1) = y^2 + 1$ Inner radius (distance from $x = -1$ to closer curve $x = 0$): $r(y) = 0 - (-1) = 1$ 4. Set up.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsExpress the following limit as a definite integral with lower limit $a = 0$, and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{2}{n}\sqrt{4 - \left(\frac{2i}{n}\right)^2}$

Identify components: $\Delta x = \frac{2}{n}$, so with $a = 0$, the upper limit is $b = 2$. Sample points are $x_i = \frac{2i}{n}$. The integrand is $f(x) = \sqrt{4 - x^2}$. The limit becomes: $\int_0^2 \sqrt{4 - x^2}\, dx$ Geometrically, $y = \sqrt{4 - x^2}$ represents the upper half of a circle of radius $r = 2$ centered at the origin. The interval $[0, 2]$ represents one quarter of the circle: $\int_0^2 \sqrt{4 - x^2}\, dx = \frac{1}{4}\pi r^2 = \frac{1}{4}\pi(2)^2 = \boldsymbol{\pi} \approx 3.1416$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 3 · AdvancedFind the coordinates of the point on the parabola $y = x^2$ that is closest to the point $(0, 3)$. Prove that this distance is a global minimum.

1. Distance formula squared: Let $(x, y) = (x, x^2)$. The squared distance is: $D(x) = (x - 0)^2 + (y - 3)^2 = x^2 + (x^2 - 3)^2 = x^2 + x^4 - 6x^2 + 9 = x^4 - 5x^2 + 9$ Minimizing $D(x)$ minimizes the actual distance $\sqrt{D(x)}$. 2. Differentiate: $D'(x) = 4x^3 - 10x = 2x(2x^2 - 5) = 0 \Rightarrow x = 0, \quad x = \pm\sqrt{\frac{5}{2}} = \pm\frac{\sqrt{10}}{2}$ 3. Second derivative: $D''(x) = 12x^2 - 10$ At $x = 0$: x and $\left(\frac{\sqrt{10}}{2}, \frac{5}{2}\right)$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 2 · MediumFind the equations of all horizontal and vertical tangent lines to the ellipse $4x^2 + 9y^2 = 36$.

Differentiate implicitly: $8x + 18y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{8x}{18y} = -\frac{4x}{9y}$. 1. Horizontal tangents (numerator = 0): $-4x = 0 \Rightarrow x = 0$. Substitute into curve: $4(0)^2 + 9y^2 = 36 \Rightarrow y^2 = 4 \Rightarrow \mathbf{y = \pm 2}$. 2. Vertical tangents (denominator = 0): $9y = 0 \Rightarrow y = 0$. Substitute into curve: $4x^2 + 9(0)^2 = 36 \Rightarrow x^2 = 9 \Rightarrow \mathbf{x = \pm 3}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 1 · FoundationsA particle moves along a line with position $s(t) = 2t^3 - 15t^2 + 24t + 10$. Determine the open intervals on $t > 0$ during which the particle is speeding up.

Find $v(t)$ and $a(t)$: $v(t) = 6t^2 - 30t + 24 = 6(t^2 - 5t + 4) = 6(t - 1)(t - 4)$ $a(t) = 12t - 30 = 6(2t - 5)$ Critical time points: $t = 1$, $t = 2.5$, $t = 4$. Construct sign chart:On $(0, 1)$: $v(t) > 0$, x: x: $v(t)0$ (opposite signs $\Rightarrow$ slowing down).On $(4, \infty)$: $v(t) > 0$, $a(t) > 0$ (same sign $\Rightarrow$ speeding up).The particle is speeding up on $(1, 2.5) \cup (4, \infty)$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumFind the particular solution to the differential equation $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(0) = -4$. State the domain of the solution.

1. Separate variables: $y\,dy = x\,dx$ 2. Integrate both sides: $\int y\,dy = \int x\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C_1 \Rightarrow y^2 = x^2 + C$ 3. Substitute initial condition $x = 0$, $y = -4$: $(-4)^2 = 0^2 + C \Rightarrow C = 16 \Rightarrow y^2 = x^2 + 16$ 4. Solve for y: Since $y(0) = -40$ for all x, y is strictly negative and never zero. The domain is $(-\infty, \infty)$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · AdvancedA population of bacteria grows at a rate proportional to its size: $\frac{dP}{dt} = kP$. At $t = 2$ hours, the population is 400. At $t = 6$ hours, the population is 1600. What was the initial population $P(0)$, and what is the doubling time?

General solution: $P(t) = P_0 e^{kt}$. From given data: $P(2) = P_0 e^{2k} = 400$ $P(6) = P_0 e^{6k} = 1600$ Divide the two equations: $\frac{P_0 e^{6k}}{P_0 e^{2k}} = \frac{1600}{400} \Rightarrow e^{4k} = 4 \Rightarrow e^{2k} = \sqrt{4} = 2$ Since $e^{2k} = 2$, the population doubles every 2 hours: Doubling time is 2 hours. To find P_0: $P_0 e^{2k} = P_0(2) = 400 \Rightarrow \mathbf{P_0 = 200}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 The Chain Rule

Level 1 · FoundationsWhich of the following functions is a solution to the differential equation $y'' + 4y = 0$? (A) $y = e^{2x}$ (B) $y = \sin(2x) - 3\cos(2x)$ (C) $y = 4\cos(4x)$ (D) $y = x^2 - 4$

Test (B): $y = \sin(2x) - 3\cos(2x)$. First derivative: $y' = 2\cos(2x) + 6\sin(2x)$ Second derivative: $y'' = -4\sin(2x) + 12\cos(2x) = -4(\sin(2x) - 3\cos(2x)) = -4y$ Substitute into the differential equation: $y'' + 4y = -4y + 4y = 0$ The equation is satisfied identically. Correct answer is (B).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Level 2 · MediumFind the value of constant k that makes $f(x) = \{kx^2 + 1, x \le 2; 3x - 5, x > 2$ continuous at $x = 2$.

Require left and right limits to match: $\lim_{x \to 2^-} (kx^2 + 1) = 4k + 1, \quad \lim_{x \to 2^+} (3x - 5) = 6 - 5 = 1.$ Set equal: $4k + 1 = 1 \Rightarrow 4k = 0 \Rightarrow \mathbf{k = 0}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

[Unit 1 · Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

[Unit 3 · Differentiation — Composite, Implicit, and Inverse Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

[Unit 4 · Contextual Applications of Differentiation · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

[Unit 5 · Analytical Applications of Differentiation · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the second derivative $f''(x)$ of $f(x) = \sin x$.

a. $-\sin x$ ✓

b. $\sin x$

c. $-\cos x$

d. $\cos x$

$f'(x) = \cos x$. Then $f''(x) = \frac{d}{dx}[\cos x] = -\sin x$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If position is $s(t) = t^4 - 2t^3 + 5$, what is the acceleration $a(t)$ at $t = 2$?

a. 24 ✓

b. 36

c. 12

d. 48

Velocity $v(t) = 4t^3 - 6t^2$. Acceleration $a(t) = 12t^2 - 12t$. At $t = 2$:
$a(2) = 12(4) - 12(2) = 48 - 24 = 24$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the 50th derivative of $f(x) = \cos x$?

a. $-\cos x$ ✓

b. $\cos x$

c. $\sin x$

d. $-\sin x$

Derivatives of cosine repeat in cycles of 4: $f^{(0)} = \cos x$, $f^{(1)} = -\sin x$, $f^{(2)} = -\cos x$, $f^{(3)} = \sin x$. Since $50 = 4(12) + 2$, $f^{(50)}(x) = f^{(2)}(x) = -\cos x$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the chain rule to find $f'(x)$ if $f(x) = (-3x - 3)^4$.

a. $4(-3x - 3)^3$

b. $-12(-3x - 3)^4$

c. $-12(-3x - 3)^3$ ✓

d. $-3(-3x - 3)^3$

Chain rule: $f'(x) = 4(-3x - 3)^3 \cdot -3 = -12(-3x - 3)^3$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.

a. **x = 3** ✓

b. x = -3 and x = 3

c. x = 1 and x = 5

d. x = 1

A horizontal tangent line occurs where the derivative of the function is equal to zero. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 6x + 10) = 2x - 6$. Next, set the derivative equal to zero and solve for x : $2x - 6 = 0$. Adding 6 to both sides gives $2x = 6$. Dividing by 2 gives $x = 3$. Therefore, the x-coordinate of the point where the tangent line is horizontal is $x = 3$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. **4** ✓

d. 8

The definite integral $\int_{-2}^{2} f(x)\,dx$ represents the net signed area between the graph of $f(x)$ and the x-axis from $x = -2$ to $x = 2$. The graph of $f(x)$ for $-2 \le x \le 2$ forms a semicircle above the x-axis with radius 2. The area of this semicircle is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi$. However, the provided figure shows a region that is a combination of triangles and a rectangle. From $x = -2$ to $x = 0$, the area is a triangle with base 2 and height 2, so the area is.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An AP Calculus BC FRQ models a population $P(t)$ satisfying the logistic differential equation $\frac{dP}{dt} = \frac{3}{600}P(600 - P)$. At what population size P is the population growing most rapidly?

a. $P = 300$ ✓

b. $P = 600$

c. $P = 150$

d. $P = 0$

In a logistic growth model dP/dt = kP(M - P), the maximum rate of growth occurs at the inflection point where P = M / 2 = 600 / 2 = 300.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

On an AP Calculus BC Free-Response Question, a particle moves along a curve with velocity vector $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. What is the speed of the particle at time $t = 1$?

a. $\sqrt{13}$ ✓

b. 5

c. $\sqrt{5}$

d. 13

Speed is the magnitude of the velocity vector: ||v(1)|| = sqrt((2*1)^2 + (3*1^2)^2) = sqrt(13).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a Taylor series BC FRQ, the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ is approximated by its first 2 terms S_2. According to the Alternating Series Error Bound, what is the maximum possible error $|S - S_2|$?

a. $\frac{1}{3^3} = \frac{1}{27}$ ✓

b. $\frac{1}{2^3} = \frac{1}{8}$

c. $\frac{1}{3^2} = \frac{1}{9}$

d. $\frac{1}{2 \cdot 3^3}$

By the Alternating Series Error Bound, |S - S_n| <= a_{n+1} = 1 / (2+1)^3 = 1 / 3^3 = 1 / 27.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let R be the region bounded by the graph of $y = \sin(\pi x)$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the x-axis. The radius of a cross-section of the solid perpendicular to the x-axis is given by $r(x) = k\sin(\pi x)$ for some positive constant k. If the volume of the solid is $\frac{\pi}{2}$ cubic units, determine the value of k.

a. 1/2

b. **1** ✓

c. 2

d. 4

The volume of a solid of revolution generated by revolving a region bounded by $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ about the x-axis is given by the disk method formula $V = \int_a^b \pi[f(x)]^2\,dx$. In this problem, the region R is bounded by $y = \sin(\pi x)$, the x-axis, $x = 0$, and $x = 1$. The radius of a cross-section is given by $r(x) = k\sin(\pi x)$. Therefore, the volume of the solid is $V = \int_0^1 \pi[r(x)]^2\,dx = \int_0^1 \pi(k\sin(\pi x))^2\,dx$. We are.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 72 ✓

b. 74

c. 144

d. 70

Each cross-section has area $y^2 = (3\sqrt{x})^2 = 9x$. $V = \int_0^4 9x\,dx = 72$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Area between $y = x^2$ and $y = x$ from 0 to 1 is:

a. $\frac{1}{3}$

b. $\frac{1}{6}$ ✓

c. $\frac{1}{2}$

d. $\frac{2}{3}$

Applying Unit 6 · Integration and Accumulation of Change principles, $\int_0^1 (x - x^2)\,dx = \frac{1}{6}$ Therefore, the verified correct option is $\frac{1}{6}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A population of bacteria, denoted by $P(t)$, grows according to the differential equation $\frac{dP}{dt} = kP$, where k is a constant representing the growth rate. It is observed that the population triples every 4 hours. If the initial population at $t = 0$ is $P_0 = 1000$, determine the population after 12 hours.

a. 10000

b. 27000

c. 81000 ✓

d. 9000

The given differential equation is $\frac{dP}{dt} = kP$. This is a first-order linear separable differential equation. We can solve it by separating variables: $\frac{dP}{P} = kdt$ Integrating both sides gives: $\int \frac{dP}{P} = \int kdt$ $\ln|P| = kt + C_1$ Exponentiating both sides: $|P| = e^{kt + C_1} = e^{C_1}e^{kt}$ Let $C = \pm e^{C_1}$. Since population must be positive, $P(t) = Ce^{kt}$. Given that the initial population at $t = 0$ is $P_0 = 1000$, we have $P(0) = Ce^{k(0)} = C$. Thus, $C = 1000$, and.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A particle's position in the first quadrant is described by a function $y(t)$, where t is time in seconds. The rate of change of its y-position with respect to time is governed by the differential equation $\frac{dy}{dt} = k\sin(t)$ for some positive constant k. The particle starts at the origin $(0, 0)$ at $t = 0$. At $t = \frac{\pi}{2}$ seconds, its y-position is $y(\frac{\pi}{2}) = 5$. \\ \\ Consider a second particle whose y-position $z(t)$ is described by the differential equation $\frac{dz}{dt} = k\cos(t)$, also starting at the origin at $t = 0$. If the first particle's y-position at $t = \frac{\pi}{4}$ is $y(\frac{\pi}{4}) = \frac{5}{\sqrt{2}}$, what is the value of k?

a. $\frac{10}{\pi}$

b. 5 ✓

c. $\frac{5\sqrt{2}}{\pi}$

d. $\frac{5}{2}$

For the first particle, we have $\frac{dy}{dt} = k\sin(t)$. Integrating both sides with respect to t: \\ $y(t) = \int k\sin(t)\,dt = -k\cos(t) + C_1$ Since the particle starts at the origin, $y(0) = 0$. \\ x So, $y(t) = -k\cos(t) + k = k(1 - \cos(t))$. We are given that $y(\frac{\pi}{2}) = 5$. \\ x Therefore, the value of the constant k is 5. Let's verify with the information about the second particle. The second particle's y-position $z(t)$ is described by $\frac{dz}{dt} = k\cos(t)$. Integrating wit.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A rectangular garden has a length that is twice its width. If the perimeter of the garden is 36 meters, what is the width of the garden?

a. **6 meters** ✓

b. 9 meters

c. 12 meters

d. 18 meters

Let the width of the garden be w meters. According to the problem, the length is twice the width, so the length is $l = 2w$ meters. The formula for the perimeter of a rectangle is $P = 2(l + w)$. We are given that the perimeter is 36 meters. So, we can set up the equation: $36 = 2(2w + w)$. Simplifying the equation: $36 = 2(3w)$ $36 = 6w$. To find the width, divide both sides by 6: $w = \frac{36}{6}$ $w = 6$. Therefore, the width of the garden is 6 meters. Common misconceptions: - Confusing length and width: If o.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

$f(x) = x^3 - 3x$. Inflection point at:

a. $x = -1$

b. $x = 3$

c. $x = 0$ ✓

d. $x = 1$

Applying Unit 5 · Analytical Applications of Differentiation principles, $f'' = 6x = 0 \Rightarrow x = 0$ Therefore, the verified correct option is $x = 0$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $\frac{dy}{dx} = \frac{x}{y}$ given $y(0) = 6$ (assume $y > 0$).

a. $y^2 + x^2 = 36$

b. y - x = 6

c. $y^2 - x^2 = 36$ ✓

d. $y^2 - x^2 = 38$

Separate: $y\,dy = x\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 - x^2 = 2C$. Using $y(0) = 6$: $2C = 6^2 = 36$, so $y^2 - x^2 = 36$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A particle moves with velocity $v(t) = 3t^2$. Displacement from $t = 0$ to $t = 2$ is:

a. **8** ✓

b. 24

c. 12

d. 16

Applying Unit 8 · Applications of Integration principles, $\int_0^2 3t^2\,dt = [t^3]_0^2 = 8$ Therefore, the verified correct option is 8.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

a. $-1 + \frac{39\sqrt{2}}{40}$

b. $\sqrt{2}$

c. $1 + \frac{39\sqrt{2}}{40}$

d. $\frac{39\sqrt{2}}{40}$ ✓

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = \sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{4}$, so $L(2 - \frac{1}{10}) \approx \frac{39\sqrt{2}}{40}$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

For $f(x) = (x - 5)(x - 9)$, determine the intervals where f is increasing and decreasing.

a. decreasing everywhere

b. increasing everywhere

c. increasing on $(-\infty, 7)$, decreasing on $(7, \infty)$

d. **decreasing on $(-\infty, 7)$, increasing on $(7, \infty)$** ✓

$f'(x) = 2x - 14 = 0$ at $x = 7$. Since f is an upward parabola, it decreases before and increases after this critical point.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

A population satisfies $\frac{dP}{dt} = \frac{1}{50}P$ with $P(0) = 390$. Find $P(100)$.

a. $P = 390 \cdot 2$

b. $P = 390e^{-2}$

c. $P = 390e^{2}$ ✓

d. $P = 390e^{2/100}$

Solving the separable DE: $P(t) = 390e^{\frac{1}{50}t}$. At $t = 100$: $P(100) = 390e^{\frac{1}{50} \cdot 100} = 390e^{2}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

If $f(x) = x^3 + 2x^2 - 6x$, is the graph of f concave up or concave down at $x = 5$?

a. cannot be determined

b. **concave up** ✓

c. concave down

d. neither

$f''(x) = 2(3x + 2)$, so $f''(5) = 34$, which is positive. Thus the graph is concave up at $x = 5$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the area of the region above the x-axis bounded by $y = -(x - 4)(x - 11)$.

a. $\frac{361}{6}$

b. $\frac{343}{6}$ ✓

c. $\frac{325}{6}$

d. $\frac{343}{3}$

The parabola crosses the x-axis at $x = 4$ and $x = 11$. Area $= \int_{4}^{11} -(x - 4)(x - 11)\,dx = \frac{343}{6}$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Area Between Curves

Find the area of the region enclosed by $y = x^2$ and $y = 7x$.

a. $\frac{355}{6}$

b. $\frac{343}{6}$ ✓

c. $\frac{331}{6}$

d. 49

Curves meet where $x^2 = 7x$, i.e. $x = 0$ or $x = 7$. Area $= \int_0^7 (7x - x^2)dx = \frac{343}{6}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Solve the initial value differential equation $\frac{dy}{dx} = \frac{4x}{y}$ with condition $y(0) = 3$ (for $y > 0$).

a. $y = \sqrt{4x^2 + 9}$ ✓

b. $y = \sqrt{4x^2 - 9}$

c. $y = 4x^2 + 3$

d. $y = \sqrt{8x^2 + 9}$

$y\,dy = 4x\,dx \Rightarrow y^2 = 4x^2 + C$. At $x = 0, y = 3 \Rightarrow C = 9$. Thus $y = \sqrt{4x^2 + 9}$.

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